

MT8002 Elements of Set Theory  
6. Cardinals without the Axiom of Choice

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# Preliminaries

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## Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Outline

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6.1 Introduction

6.2 Comparing Sizes

6.3 Infinite Sets without AC - Countable Sets

6.4 Uncountable Sets and Cardinal Arithmetic, without AC

References

# Outline

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## 6.1 Introduction

## 6.2 Comparing Sizes

## 6.3 Infinite Sets without AC - Countable Sets

## 6.4 Uncountable Sets and Cardinal Arithmetic, without AC

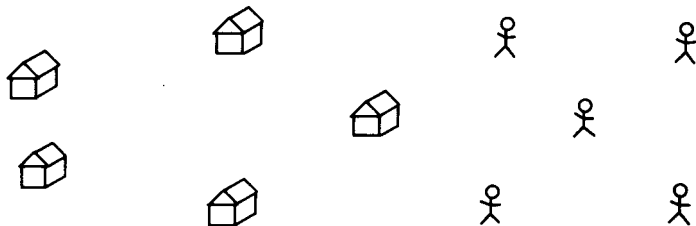
## References

# Introduction

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## Question

Suppose you only know how to count to three. Are the set of houses and the set of people in the figure the same size?<sup>†</sup>



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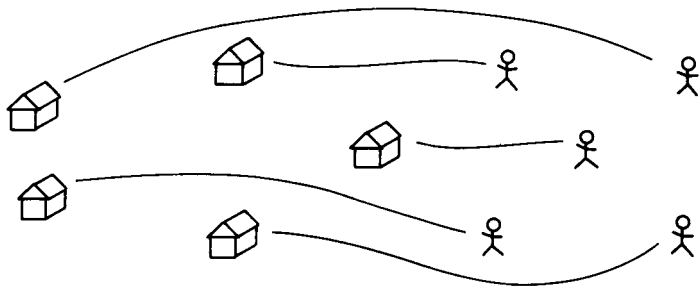
<sup>†</sup>Figure source: (Enderton 1977, Fig. 30).

# Introduction

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## Answer

Yes! The set of houses is the same size as the set of people, as shown in the figure.<sup>†</sup>



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<sup>†</sup>Figure source (Enderton 1977, Fig. 31).

# Introduction

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## Question

In the previous question, the sets were finite. Can we apply similar reasoning if the sets are infinite?

# Introduction

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## Definition (p. 127)

A set  $X$  is **finite** if there is a bijective function  $f : \mathbf{n} \longrightarrow X$ , for some natural number  $\mathbf{n}$ .

## Definition (p. 127)

A set  $X$  is **infinite** if it is not finite. That is, there is no a bijective function  $f : \mathbf{n} \longrightarrow X$ , for any natural number  $\mathbf{n}$ .



# Outline

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6.1 Introduction

**6.2 Comparing Sizes**

6.3 Infinite Sets without AC - Countable Sets

6.4 Uncountable Sets and Cardinal Arithmetic, without AC

References

# Comparing Sizes

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## Definition (p. 130)

A set  $A$  is **equinumerous** with a set  $B$ , denoted  $A \approx B$ , if there is a bijective function  $f : A \longrightarrow B$ .

# Comparing Sizes

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## Definition (p. 130)

A set  $A$  is **equinumerous** with a set  $B$ , denoted  $A \approx B$ , if there is a bijective function  $f : A \longrightarrow B$ .

## Definition (p. 130)

A set  $A$  is **less than or equinumerous** with a set  $B$ , denoted  $A \lesssim B$ , if there is a one-to-one function  $f : A \longrightarrow B$ .

# Comparing Sizes

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## Definition (p. 130)

A set  $A$  is **equinumerous** with a set  $B$ , denoted  $A \approx B$ , if there is a bijective function  $f : A \longrightarrow B$ .

## Definition (p. 130)

A set  $A$  is **less than or equinumerous** with a set  $B$ , denoted  $A \lesssim B$ , if there is a one-to-one function  $f : A \longrightarrow B$ .

## Definition (p. 130)

A set  $A$  is **dominated** by a set  $B$ , denoted  $A < B$ , if  $A \lesssim B$  and it is not the case that  $A \approx B$ .

# Comparing Sizes

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## Notation

$$X \not\approx Y := \neg(X \approx Y),$$

$$X \not\lesssim Y := \neg(X \lesssim Y),$$

$$X \not< Y := \neg(X < Y).$$

# Comparing Sizes

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## Examples

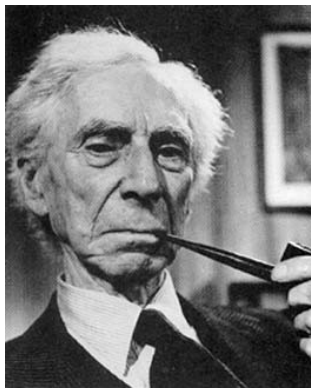
(a)  $\mathbb{N} \lesssim \mathbb{Z}$  and  $\mathbb{Z} \lesssim \mathbb{N}$ .

(b)  $\mathbb{N} \approx \mathbb{Z}$ .

(c)  $\mathbb{N} \not\lesssim \mathbb{Z}$  and  $\mathbb{Z} \not\lesssim \mathbb{N}$ .

# Comparing Sizes

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(1872 – 1970)

*“The possibility that whole and part may have the same number of terms is, it must be confessed, shocking to common-sense.” (Russell 1903, p. 358)*

# Outline

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6.1 Introduction

6.2 Comparing Sizes

6.3 Infinite Sets without AC - Countable Sets

6.4 Uncountable Sets and Cardinal Arithmetic, without AC

References



## Infinite Sets without AC - Countable Sets

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### Definition (p. 143)

A set  $A$  is **countable** if it is finite or equinumerous with  $\mathbb{N}$ .

### Definition (p. 143)

A set  $A$  is **countably infinite** if it is equinumerous with  $\mathbb{N}$ .

# Infinite Sets without AC - Countable Sets

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## Examples

- Let  $n \in \mathbb{N}$ . The set  $n$  is countable because it has  $n$  elements.
- The sets  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{N} \times \mathbb{N}$  and  $\mathbb{Q}$  are countably infinite.

# Infinite Sets without AC - Countable Sets

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## Theorem 6.5 (Cantor's Theorem)

For any set  $X$ ,  $X < \mathcal{P}(X)$ .

### Example

$$\mathbb{N} < \mathcal{P}(\mathbb{N}) < \mathcal{P}(\mathcal{P}(\mathbb{N})) < \dots$$

## Infinite Sets without AC - Countable Sets

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### Definition (p. 148)

A set **uncountable** or **uncountably infinite** is an infinite set which is not countable.

# Outline

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6.1 Introduction

6.2 Comparing Sizes

6.3 Infinite Sets without AC - Countable Sets

6.4 Uncountable Sets and Cardinal Arithmetic, without AC

References

# Uncountable Sets and Cardinal Arithmetic, without AC

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## Remark

Recall that  $2^X \doteq \{0,1\}^X$  is the set of functions from the set  $X$  to  $\{0,1\}$ .

# Uncountable Sets and Cardinal Arithmetic, without AC

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## Remark

Recall that  $2^X \doteq \{0,1\}^X$  is the set of functions from the set  $X$  to  $\{0,1\}$ .

## Theorem 6.7

For any set  $X$ ,  $\mathcal{P}(X) \approx 2^X$ .

## Theorem 6.8

The set  $\mathbb{R}$  is uncountable.

## Theorem

$\mathbb{R} \approx \mathcal{P}(\mathbb{N}) \approx 2^{\mathbb{N}}$ .

# Outline

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6.1 Introduction

6.2 Comparing Sizes

6.3 Infinite Sets without AC - Countable Sets

6.4 Uncountable Sets and Cardinal Arithmetic, without AC

References



# References

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Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on pp. 5, 6).

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).

Bertrand Russell (1903). The Principles of Mathematics. Cambridge University Press (cit. on p. 15).